Dominick Robinson

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EDUCATION

Carnegie Mellon University **Expected May 2026**
Master of Science in Applied Data Science (MADS) *Pittsburgh, PA*
Rales Fellow
Highly selective graduate program dedicated to cultivating the next generation of STEM leaders through faculty mentorship, professional development, and networking opportunities with academic and industry leaders
Relevant coursework: Applied Linear Models, Statistical Principles of Generative AI, Statistical Computing, Data Visualization, Data Engineering & Distributed Environments, Software for Large-Scale Data, Computational Methods for Statistics, Time Series, Statistical Machine Learning, Statistical Practice

Carnegie Mellon University **May 2025**
Bachelor of Science in Mathematical Sciences (Statistics), Minor in Creative Writing *Pittsburgh. PA*
3.85/4.00 GPA
Relevant coursework: Matrices and Linear Transformations, Principles of Imperative Computation, Multidimensional Calculus, Probability, Principles of Real Analysis I, Intro to Statistical Inference II, Intro to Ordinary Differential Equations, Discrete Math, Numerical Methods, Operations Research I & II, Functional Programming, Intro to Probability Models, Text Analysis, Parallel and Sequential Data Structures and Algorithms, Advanced Data Analysis, Principles of Software Construction

SKILLS

Programming: Python, C, HTML, CSS, Pandas, Dash, R, Plotly, SQL, Streamlit, Matplotlib, Beautiful Soup, JavaScript, Java, TypeScript
Tools: LaTeX, GitHub, Git, VSCode, PowerPoint, Excel, Word, Godot
Functional: Communication, Leadership, Adaptability, Creativity

TECHNICAL PROJECTS

Applied Linear Models Final Project: HIV Treatment Report **Nov 2025 - Dec 2025**
- Analyzed clinical trial data to assess HIV therapy effectiveness, finding baseline CD4 count and prior therapy duration predict short-term response, but not long-term immunologic outcomes

Data Engineering Final Project: Automated Data Pipeline and Dashboard **Nov 2025 - Dec 2025**
- Collaborated with team to design PostgreSQL schema and ETL pipeline for College Scorecard and IPEDS data
- Developed interactive dashboard with team to visualize and query College Scorecard and IPEDS data

Applied Linear Models Midterm Project: Rail Trails Report **Oct 2025**
- Investigated impact of proximity to rail trails on home values, finding no significant effect

Data Visualization Final Project: Dominance of the Big 3/4 in Men's Tennis **Oct 2025**
- Cleaned and analyzed ATP Tennis data in R to visualize Big 3/4 dominance in men's tennis

Text Analysis Final Project: Exploring Linguistic Differences Across Generations **Nov 2024 - Dec 2024**
- Scraped, cleaned, and tokenized thousands of Reddit comments in Python to build a generational dataset
- Analyzed Gen Alpha, Gen Z, and Millennial text, uncovering shifts in slang, sentiment, and topic use

Operations Research II Final Project: Feeding CMU Orientation **Oct 2024 - Dec 2024**
- Scraped and cleaned nutritional information and prices of Trader Joe's and Whole Foods products using Python
- Mapped products to ingredients in recipes database using Fuzzy library
- Solved integer program to minimize cost of feeding 900 CMU first-years with dietary and nutritional constraints

Tennis ELO Dashboard **Jan 2024**
- Used Plotly Dash to model ATP tennis ELO and what-if scenarios over time, calculated from tennis dataset

Modern Regression Term Projects **Oct 2023 - Nov 2023**

- Used R to analyze the impact of sleep on GPA using linear regression, revealing significant correlations
- Developed predictive model by analyzing instructor evaluation data to identify biases affecting quality scores, revealing significant impacts from course difficulty, instructor attractiveness, gender, and discipline

Intro to Machine Learning Projects **July 2023**

- Used Python to develop a neural network from scratch for optimizing optical handwritten letter classification with stochastic gradient descent
- Developed an HMM-based Named Entity Recognition system, utilizing forward-backward algorithm and log-space arithmetic for efficient entity classification with data

Graph Theory Research **May 2023 - Dec 2023**

- Developed mathematical tool based on the graph theory game of "cops and robbers"
- Designed graph editor, JSON converter/parser, and computer opponents with optimal strategies

Gamification in the classroom: How different types of leaderboards and levels of goal-setting affect performance and motivation in a problem solving task **Sept 2020 - May 2021**

- Developed live online slide puzzle software using Python
- Collected and analyzed various statistics of 100+ students using Google Sheets API

WORK EXPERIENCE

RedZone Robotics **July 2024 - Aug 2025**

Engineering Intern

- Used Python and OpenCV to digitize images of lasers for camera calibration
- Developed software and assess various optimization algorithms for autonomously calibrating fisheye cameras

Mathematical Sciences Department, CMU **Aug 2023 - May 2024**

Teaching Assistant for Multivariate Analysis

- Created and delivered lesson plans for weekly recitations of 30 students
- Hosted office hours, graded homeworks and exams, and created homework solutions using LaTeX

LEADERSHIP

Game Creation Society Project (5 Nights on Wean 4) **Sept 2025 - Dec 2025**

Team Lead

Game Creation Society Project (GCS Smash) **Sept 2024 - Dec 2024**

Team Lead

CMU Pickleball Club, President/Founder **Dec 2022 - Dec 2024**

- Grew club to 500+ members over 4 semesters
- Developed club website (cmupickleball.net)

Game Creation Society, President **Dec 2022 - Dec 2023**

- Communicated and planned talks with industry speakers

Game Creation Society Project (Mad Rabbits: Epic Vegetable Quest) **Sept 2022 - Dec 2022**

Team Lead

Game Creation Society Project (Supreme Hen Donkey), Team Lead **Feb 2022 - May 2022**

- Quickly learned game development in Godot with no prior experience to deliver quality product
- Won **GCS Gold Award** for excellence